

SECURITY OPERATIONS — INCIDENT RESPONSE

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# Incident Report:

## DNS Resolution Failure Analysis

Network traffic analysis of a DNS service outage affecting  
www.yummyrecipesforme.com

## INCIDENT ID

INC-2026-0142

## SEVERITY

Medium

## DATE OF REPORT

August 11, 2026

## STATUS

Under Investigation

## PREPARED BY

Security Analyst

## ANALYSIS TOOL

tcpdump (packet capture)

## AFFECTED SERVICE

Domain Name System (DNS)

## AFFECTED HOST

www.yummyrecipesforme.com

## 1. Executive Summary

Multiple users reported being unable to access **www.yummyrecipesforme.com**, receiving the error “**destination port unreachable**.” The issue was reproduced and investigated using the network protocol analyzer **tcpdump**.

The packet capture showed that the browser sent DNS queries to the DNS server using **UDP on port 53**. Instead of a DNS response, the client received **ICMP** messages indicating that **UDP port 53 was unreachable**.

The incident affected the **DNS service** and prevented the domain name from being resolved to an IP address, which in turn prevented the browser from establishing the subsequent HTTPS connection.

**Scope note:** the packet capture identifies the immediate technical problem but does not establish the underlying root cause. Additional investigation of the DNS server, firewall, system logs, configuration, and administrative activity is required.

## 2. Incident Description

The company received reports from several customers who were unable to access **www.yummyrecipesforme.com**. When attempting to access the website, users received the error “**Destination Port Unreachable**.” To investigate, the analyst attempted to access the website while capturing network traffic with **tcpdump**.

### Normal DNS resolution process

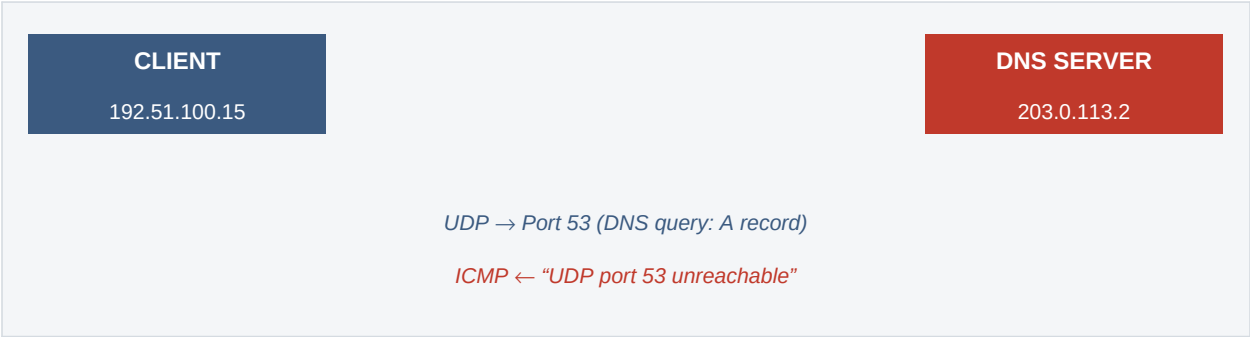
- The browser requests the IP address associated with **www.yummyrecipesforme.com**.
- The request is sent to a DNS server.
- The DNS server resolves the domain name to an IP address.
- The browser uses the resulting IP address to establish an HTTPS connection with the web server.

In this incident, the **DNS resolution step failed**.

## 3. Network Traffic Analysis

The **tcpdump** capture showed that the browser sent DNS queries using UDP to the DNS server. The initial request contained the parameters below.

|                  |                                                  |
|------------------|--------------------------------------------------|
| Source IP        | 192.51.100.15                                    |
| Destination IP   | 203.0.113.2                                      |
| Protocol         | UDP                                              |
| Destination port | 53                                               |
| DNS query        | www.yummyrecipesforme.com                        |
| Query type       | A record (maps a domain name to an IPv4 address) |



#### 4. DNS Failure

Instead of returning a DNS response, the server sent an **ICMP** response indicating “**udp port 53 unreachable.**” The ICMP response contained the parameters below.

|                |               |
|----------------|---------------|
| Source IP      | 203.0.113.2   |
| Destination IP | 192.51.100.15 |
| Protocol       | ICMP          |

The error indicates that the UDP packet could not be delivered to port 53 because **no service was listening on the receiving DNS port**. The same delivery error occurred multiple times.

#### 5. Affected Protocol and Service

|                                     |                                          |
|-------------------------------------|------------------------------------------|
| Primary affected service            | Domain Name System (DNS)                 |
| Protocol used for the DNS request   | User Datagram Protocol (UDP)             |
| Destination port                    | UDP 53                                   |
| Protocol used to report the failure | Internet Control Message Protocol (ICMP) |

The incident did not indicate that HTTPS itself was malfunctioning. The failure occurred earlier in the communication process, during DNS resolution.

#### 6. Impact

Because the DNS server was unable to respond to requests on UDP port 53, clients could not obtain the IP address associated with `www.yummyrecipesforme.com`. Without the destination IP address, the browser could not proceed to establish the HTTPS connection required to display the website. The incident therefore resulted in a

website availability issue caused by DNS resolution failure.

## 7. Findings

Based on the available packet capture, the following findings were established:

| Finding                          | Result                     |
|----------------------------------|----------------------------|
| DNS query transmitted by client  | Confirmed                  |
| DNS protocol involved            | Confirmed                  |
| UDP used for DNS request         | Confirmed                  |
| Destination port                 | UDP 53                     |
| DNS response received            | No                         |
| ICMP error received              | Confirmed                  |
| Error                            | UDP port 53 unreachable    |
| DNS service available on port 53 | No                         |
| HTTPS connection possible        | No — DNS resolution failed |
| Root cause of DNS unavailability | Not determined             |

## 8. Root Cause Assessment

The network capture establishes that the DNS service was unavailable on **UDP port 53** at the time of the incident. However, the available packet data does **not** establish why the service was unavailable.

**The root cause could not be determined from the network capture alone.** Further investigation is required before attributing the incident to a particular operational or security cause.

## 9. Additional Investigation

The following steps are recommended to determine the underlying cause of the DNS service failure.

### 9.1 Verify DNS Service Status

First, verify whether the DNS service is currently running on the DNS server.

- Whether the DNS process is active
- Whether the service stopped unexpectedly
- Whether restarting the service resolves the issue
- Whether the service reports startup or runtime errors

This would help determine whether the problem was caused by a service failure rather than a network problem.

### 9.2 Review DNS and System Logs

The DNS server's application and operating-system logs should be reviewed around the time of the incident. The analyst should look for:

- Service crashes
- Unexpected shutdowns
- Configuration errors
- Resource exhaustion
- Startup failures
- Other events occurring immediately before the outage

The goal is to establish when the DNS service stopped responding and what happened immediately beforehand.

### 9.3 Inspect Firewall Configuration

The firewall configuration should be reviewed to determine whether UDP traffic on port 53 is being permitted. The investigation should include:

- Current firewall rules
- Recently modified rules
- Blocked UDP/53 traffic
- Changes made around the time of the incident

A firewall configuration problem could prevent DNS communication even if the DNS service itself is functioning correctly.

### 9.4 Validate DNS Configuration

The DNS server configuration should be examined to verify that the service is configured to listen for DNS requests on UDP port 53. Recent configuration changes should also be reviewed.

This is important because a DNS service can be running while still being incorrectly configured or listening on an unexpected interface or port.

### 9.5 Verify Network Connectivity

Connectivity between clients and the DNS server should be tested. The investigation should determine whether:

- The DNS server is reachable
- Routing is functioning correctly
- Packets are reaching the server
- Responses are able to return to clients

This helps distinguish a DNS service problem from a broader network connectivity problem.

### 9.6 Review Recent System Changes

Recent changes to the environment should be identified, including:

- Software updates
- Operating-system updates
- DNS configuration changes
- Firewall changes
- Maintenance activities
- Infrastructure modifications

If the timing of one of these changes corresponds with the beginning of the outage, it may provide an important lead toward identifying the root cause.

### 9.7 Review Authentication and Administrative Activity

Administrative and authentication logs should be reviewed to determine whether privileged accounts accessed the DNS server shortly before the incident. The investigation should look for:

- Unexpected administrator logins
- Unusual authentication activity
- Changes made by privileged accounts
- Modifications to DNS configuration
- Modifications to firewall configuration

This step does not assume that the incident was caused by malicious activity. It is intended to determine whether unauthorized or unexpected administrative activity contributed to the outage.

## 9.8 Investigate Potential Security Causes if Evidence Supports Them

If the previous investigation produces evidence suggesting malicious activity, additional security investigation should be performed. Potential causes could include:

- Unauthorized modification of DNS configuration
- Unauthorized firewall changes
- Deliberate termination of the DNS service
- Malware affecting the DNS server
- Denial-of-service activity

At this stage, these possibilities should be treated as hypotheses rather than conclusions. If compromise is suspected, relevant logs and forensic evidence should be preserved before making significant changes to the affected system.

## 10. Role of Network Analysis Tools

The tcpdump capture was useful for identifying the network-level symptom of the incident. It allowed the analyst to determine which protocol was being used, which IP addresses were communicating, which port was being targeted, what response was returned, and what error occurred.

However, tcpdump only provides visibility into **network traffic**. It cannot, by itself, determine whether the DNS service was stopped, whether a firewall rule was changed, whether a configuration file was modified, or whether an administrator accessed the server. Further investigation requires information from additional sources, such as:

- DNS logs
- Operating-system logs
- Firewall logs
- Authentication logs
- Configuration records
- Security monitoring systems

A **SIEM** could be useful for centralizing and correlating these different sources of information during the investigation.

## 11. Conclusion

The investigation confirmed that the incident was caused by a failure in the **DNS resolution process**. The client successfully sent DNS requests using UDP to port 53, but instead of receiving a DNS response, it received ICMP “UDP Port 53 Unreachable” messages. As a result, the domain name could not be resolved to an IP address, preventing the browser from establishing the subsequent HTTPS connection to the website.

The available network traffic identifies the immediate technical issue but does not establish its root cause. The next stage of the investigation should therefore focus on the DNS server's service status, system and DNS logs, firewall configuration, DNS configuration, network connectivity, recent changes, and administrative activity. Only after

these sources of evidence have been reviewed should a definitive root cause be assigned to the incident.